

Dorsa Dolatkahh Ouchbolagh

Polymer Engineer — M.Sc. Candidate, University of Tehran

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EDUCATION

B.Sc. in Polymer Engineering

2022 – 2026

School of Polymer Engineering, Department of Chemical Engineering, College of Engineering, University of Tehran

GPA: 4.00 / 4.00 (18.04 / 20)

RESEARCH INTERESTS

Sustainable & Biodegradable Polymers · AI and Machine Learning in Polymer Science · Polymer Recycling · Smart and Stimuli-Responsive Biosensors · Biomedical Polymers · Bio-based Polymer Composites · Bio-inspired materials

SELECTED COURSEWORK

- Rubber and Elastomer Engineering 18.0/20
- Physical and Mechanical Properties of Polymers 19.92/20
- Kinetics and Synthesis of Polymers 18.5/20
- Polymers and the Environment 20.0/20
- Polymer Molecular Analysis 19.5/20
- Polymer Rheology 18.6/20
- Physical Chemistry of Polymers 20.0/20
- Resins Engineering 19.6/20
- Natural Polymers 20.0/20
- Principles of Polymerization 19.8/20

RESEARCH and PROJECTS

- Bachelor's thesis: a review of bio-based phase change materials for thermal and energy management, with applications in building structures, batteries, and personal thermal-management patches.
- Designed and Fabricated bagasse-based food containers embedded with phase change materials to control heat dissipation and cold storage.
- A Class project on smart, sustainable and biodegradable polymers (e.g., PLA) using 4D filament printing with heat sensitivity for shape memory and reversibility properties.

PUBLICATION

- A Review on Biomass Selection for Phase Change Material-Embedded Composites: Criteria, Material Properties, and Design Considerations. [in Preparation]

EXPERIENCE

Teaching Assistant — Calculus I & II

Fall 2022

Department of Chemical Engineering, University of Tehran

- Assisted the course instructor in grading assignments and supporting classroom activities.

Teaching Assistant — Differential Equations (ODE)

Spring 2023

Department of Chemical Engineering, University of Tehran · Course instructor: Prof. H. Rahami

- Guided undergraduate students during tutorials and problem-solving sessions.
- Communicated complex engineering concepts in a clear and accessible manner.

Laboratory Intern

Summer 2025

Rubber Research and Engineering Company (VardAvard), Tehran, Iran · 2-month internship

- Contributed to interdisciplinary research projects supporting rubber and automotive industry applications.
- Senior Assistance in Polymer processing laboratory and rheological, physical and mechanical test procedures to ensure accurate experimental procedures and properties based on ASTM standards.

SKILLS

Technical Skills

Technical Analysis

MATLAB, Python, Aspen Plus

Rheometry, DSC, XRD, TGA, DMTA

AWARDS AND HONORS

- Ranked 1st in graduating class of the B.Sc. Polymer Engineering program, School of Polymer Engineering, University of Tehran.
- Admitted to M.Sc. programs at the University of Tehran, Amirkabir University of Technology, and Sharif University of Technology without entrance examination, on the basis of academic excellence.

LANGUAGE SKILLS

Persian

English

Native

Advanced — TOEFL iBT, Summer 2026: [score pending]; Certificate of Proficiency in English, Bloomsbury International (UK), 2018: 89/100

REFERENCES

Dr. SeyedMostaffa Hosseini — Assistant Professor, Department of Polymer Engineering, School of Chemical Engineering, College of Engineering, University of Tehran

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Dr. Sara Tarashi — Assistant Professor, Department of Polymer Engineering, School of Chemical Engineering, College of Engineering, University of Tehran

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